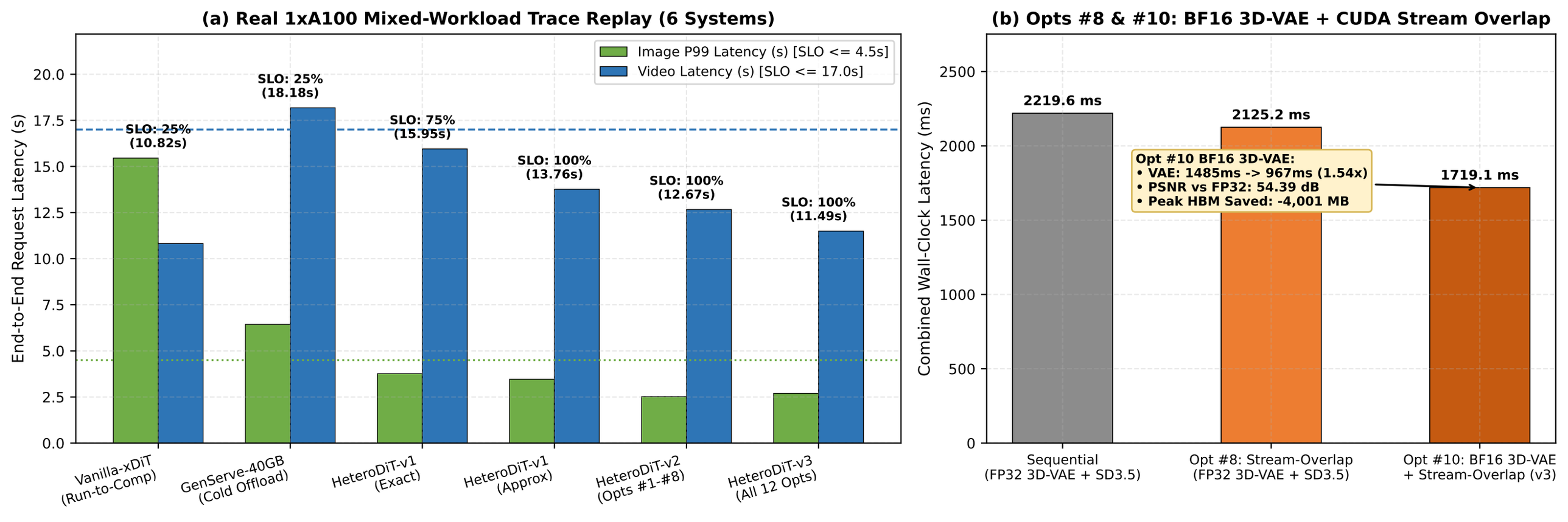
(a) Real 1xA100 Mixed-Workload Trace Replay (6 Systems)
End-to-End Request Latency (s)
Image P99 Latency (s) [SLO <= 4.5s]
Video Latency (s) [SLO <= 17.0s]
SLO: 25% (10.82s)
SLO: 25% (18.18s)
SLO: 75% (15.95s)
SLO: 100% (13.76s)
SLO: 100% (12.67s)
SLO: 100% (11.49s)
Vanilla-xDiT (Run-to-Comp)
GenServe-40GB (Cold Offload)
HeteroDiT-v1 (Exact)
HeteroDiT-v1 (Approx)
HeteroDiT-v2 (Opts #1-#8)
HeteroDiT-v3 (All 12 Opts)
(b) Opts #8 & #10: BF16 3D-VAE + CUDA Stream Overlap
Combined Wall-Clock Latency (ms)
2219.6 ms
2125.2 ms
1719.1 ms
Opt #10 BF16 3D-VAE:
• VAE: 1485ms -> 967ms (1.54x)
• PSNR vs FP32: 54.39 dB
• Peak HBM Saved: -4,001 MB
Sequential (FP32 3D-VAE + SD3.5)
Opt #8: Stream-Overlap (FP32 3D-VAE + SD3.5)
Opt #10: BF16 3D-VAE + Stream-Overlap (v3)